

The Importance of PCI-DSS

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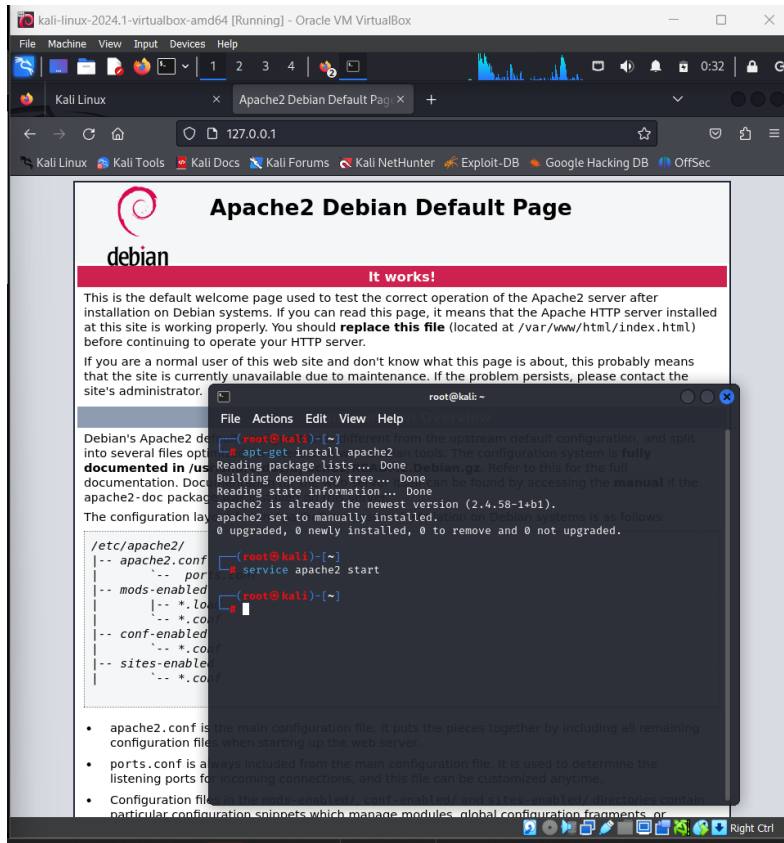
Grand Canyon University, College of Engineering and Technology

Professor Christopher Molstad

4/15/2024

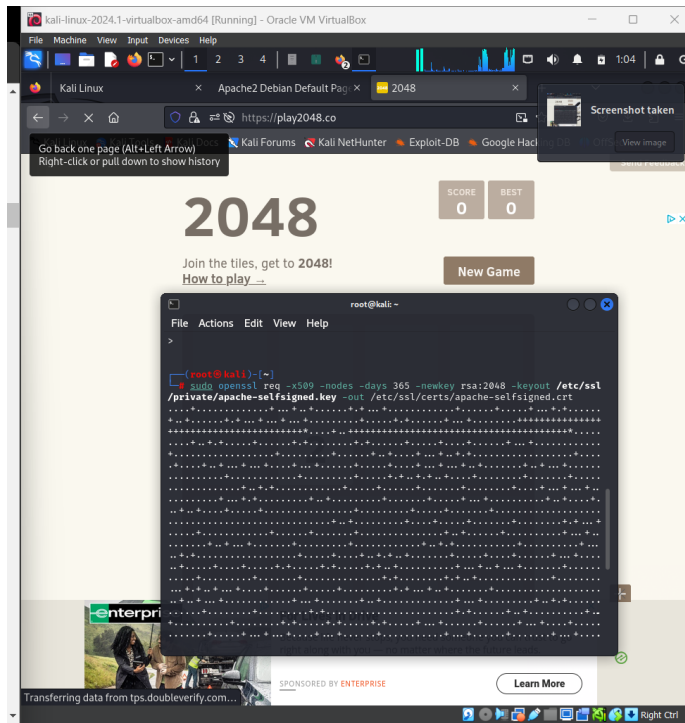
It is necessary for all organizations who store, process, transfer credit card data information to comply with standards and practices set by PCI-DSS. The Payment Card Industry Data Security Standards are essential to application both in operational and technical settings. The overarching goal of these standards is to protect the cardholder's data. There are twelve main requirements listed in the PCI DSS, such as installing and maintaining firewall configurations, not relying on system/vendor-supplied defaults for passwords and parameters, implement secure measures for stored data, encrypt all transmission especially if it is transmitting cardholder data across public networks. Restrict the amount of access to cardholder data by a need to know basis, restrict physical access to equipment, paperwork, systems that can hold or process cardholder data, and place procedural tracking/monitoring of accessibility to network resources and data. Ensure that there are regular testings of security systems and processes in place and maintain a policy that addresses information security for all personnel. If an organization fails to comply with these standards they not only put their organization in jeopardy but they risk exposing a list of personal identifiable information of those whose credit card information is being stored, processed or transmitted. In addition to having technical processes in place it is imperative that all users/staff members in the organization understand that their everyday actions and awareness can be on step closer to security or opening a vulnerability to malicious actors looking to gain access to this information. If a actor is able to obtain cardholder data, they can utilize these credentials to commit identity theft, fraud, sell on the darknet, hold for ransom, and more. Ensure that all users who have access to your organization's systems are authorized, have limited access, and pass background checks.

Started by downloading and starting the apache server.

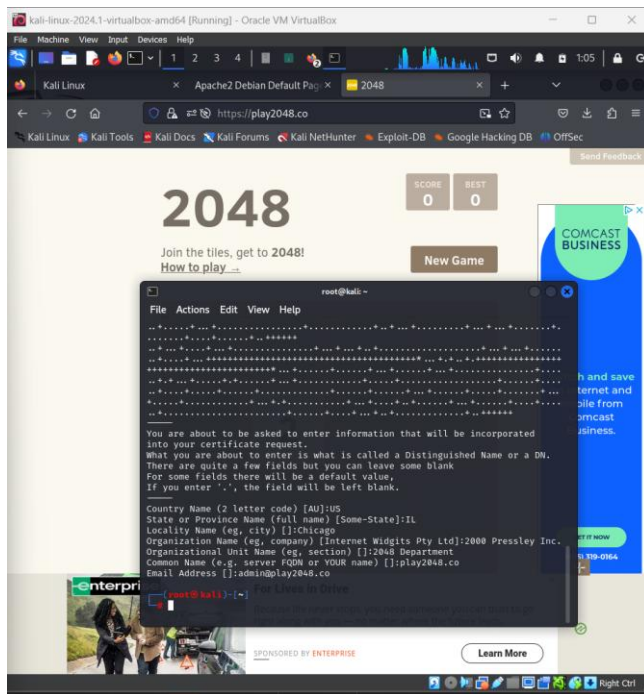


I didn't have access or idea for vulnerable web applications. I could test this one so I picked a random web application- the game 2048. After downloading open ssl and apache2, I opened 2048 on the browser. Though the web application is already secure I was still able to understand the process necessary to enable ssl security on a web application.

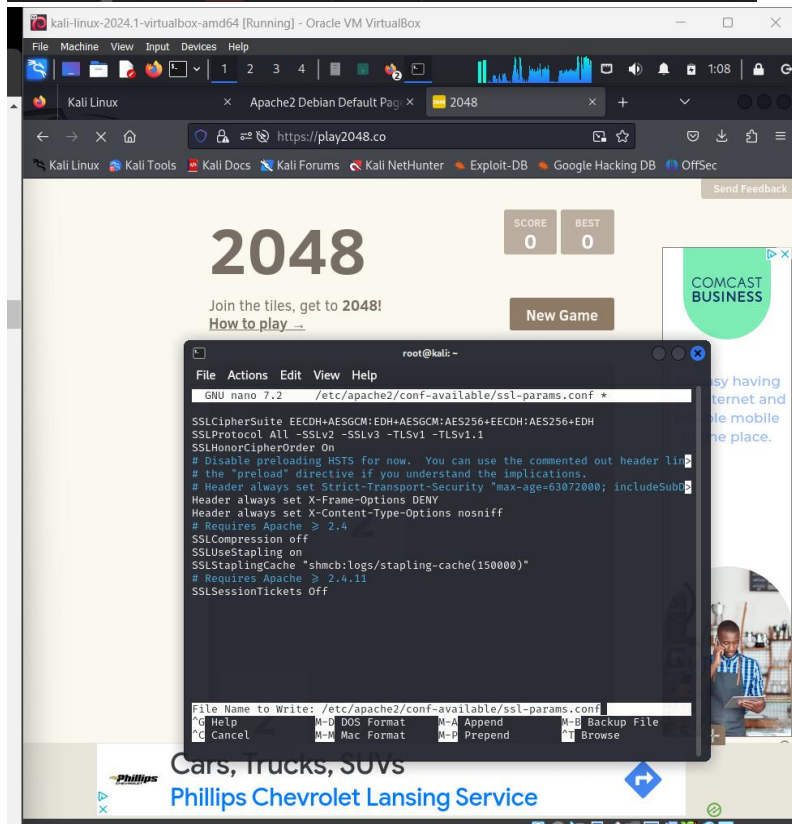
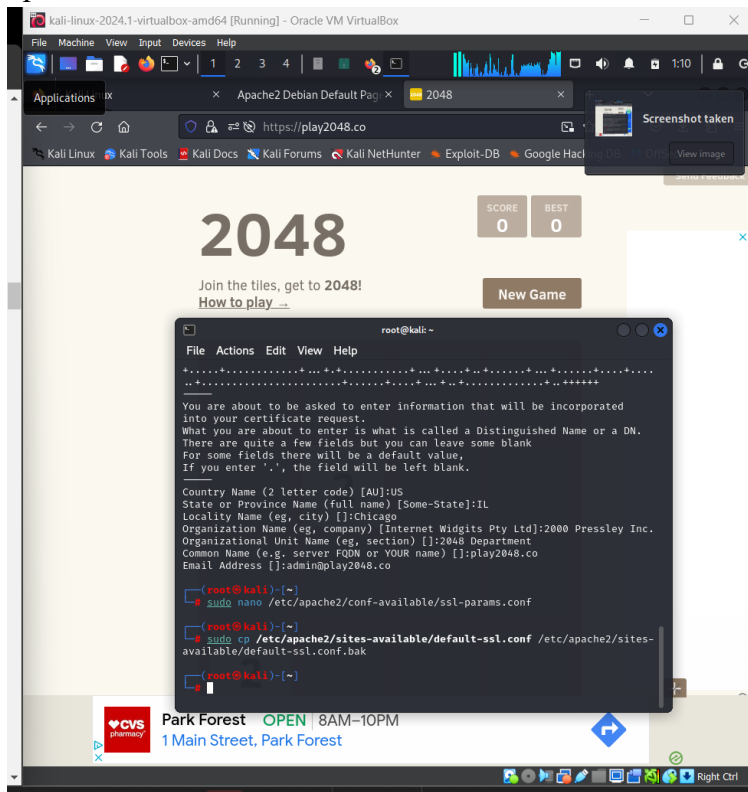
I then entered a command which essentially creates a self-signed key and certificate pair with OpenSSL.



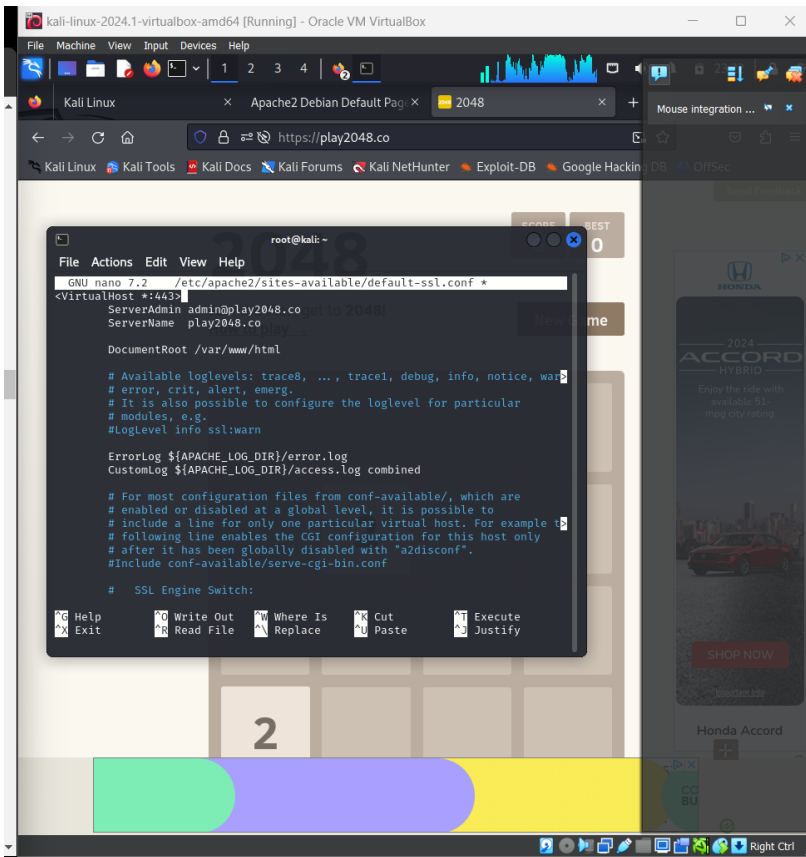
Afterwards it asks a series of questions such as the location, domain name, server name and more.



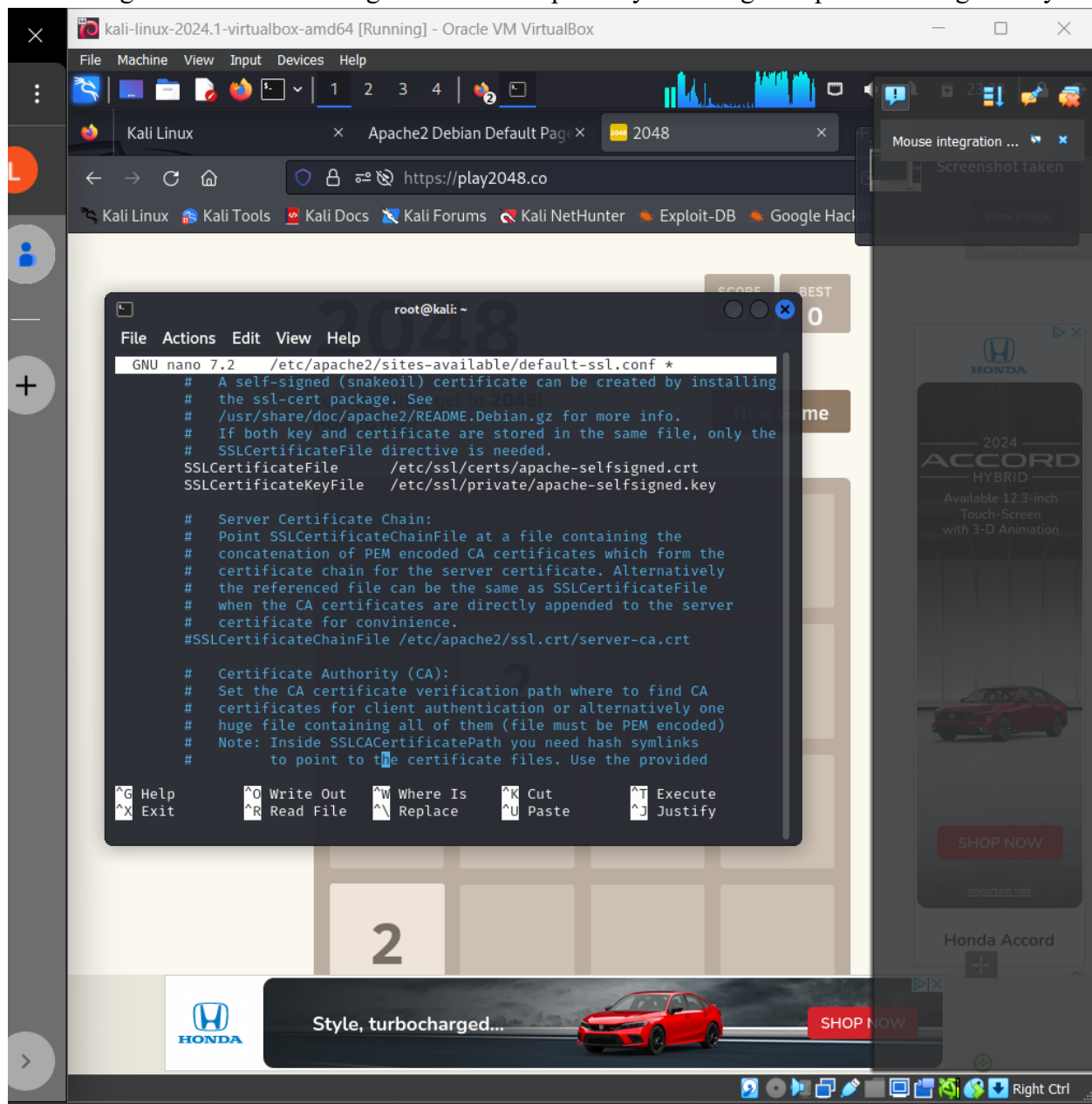
I then enter a nano command to open the apache file that contains the configuration for the openssl.



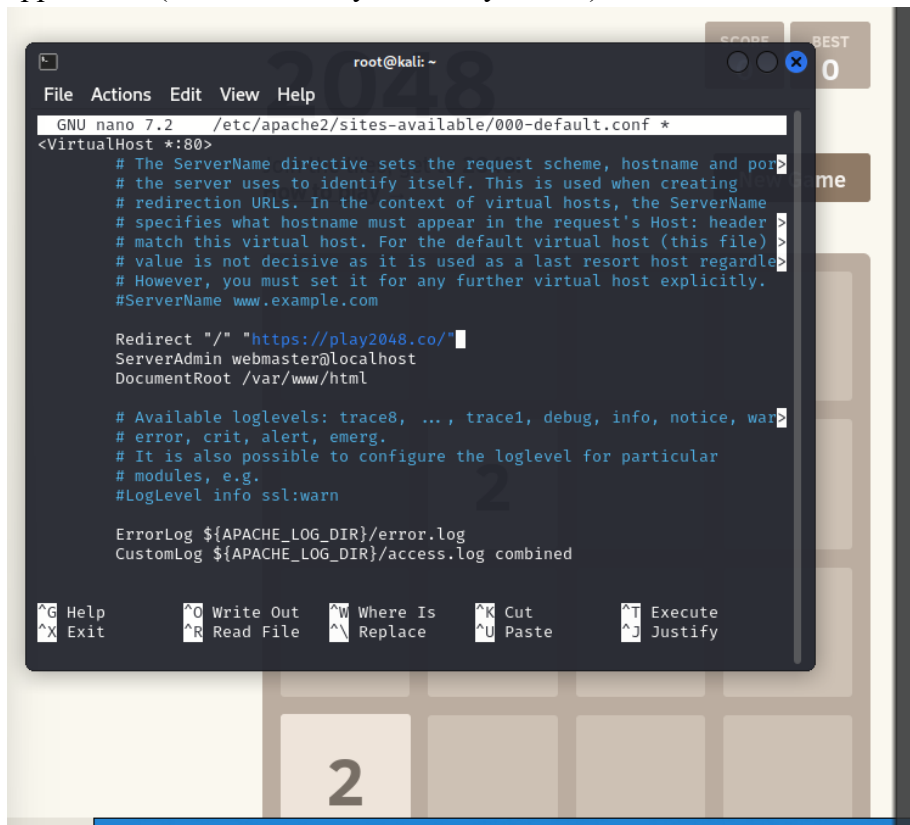
Then using another nano command we changed the server admin to the admin name that associates with 2048 along with the domain name. And changed the errorlog and customlog to apache logs.



Also changed in that same configuration the file pathways to being the apache self-signed keys.



Finally I was able to change the redirection of the domain name to the secured version of the web application (which in reality is already secure).



The screenshot shows a terminal window with a nano editor editing the file `/etc/apache2/sites-available/000-default.conf`. The user is `root@kali`. The configuration file content is as follows:

```
GNU nano 7.2 /etc/apache2/sites-available/000-default.conf *
<VirtualHost *:80>
    # The ServerName directive sets the request scheme, hostname and port
    # the server uses to identify itself. This is used when creating
    # redirection URLs. In the context of virtual hosts, the ServerName
    # specifies what hostname must appear in the request's Host: header
    # match this virtual host. For the default virtual host (this file)
    # value is not decisive as it is used as a last resort host regardless
    # However, you must set it for any further virtual host explicitly.
    #ServerName www.example.com

    Redirect "/" "https://play2048.co/"
    ServerAdmin webmaster@localhost
    DocumentRoot /var/www/html

    # Available loglevels: trace8, ..., trace1, debug, info, notice, warn
    # error, crit, alert, emerg.
    # It is also possible to configure the loglevel for particular
    # modules, e.g.
    #LogLevel info ssl:warn

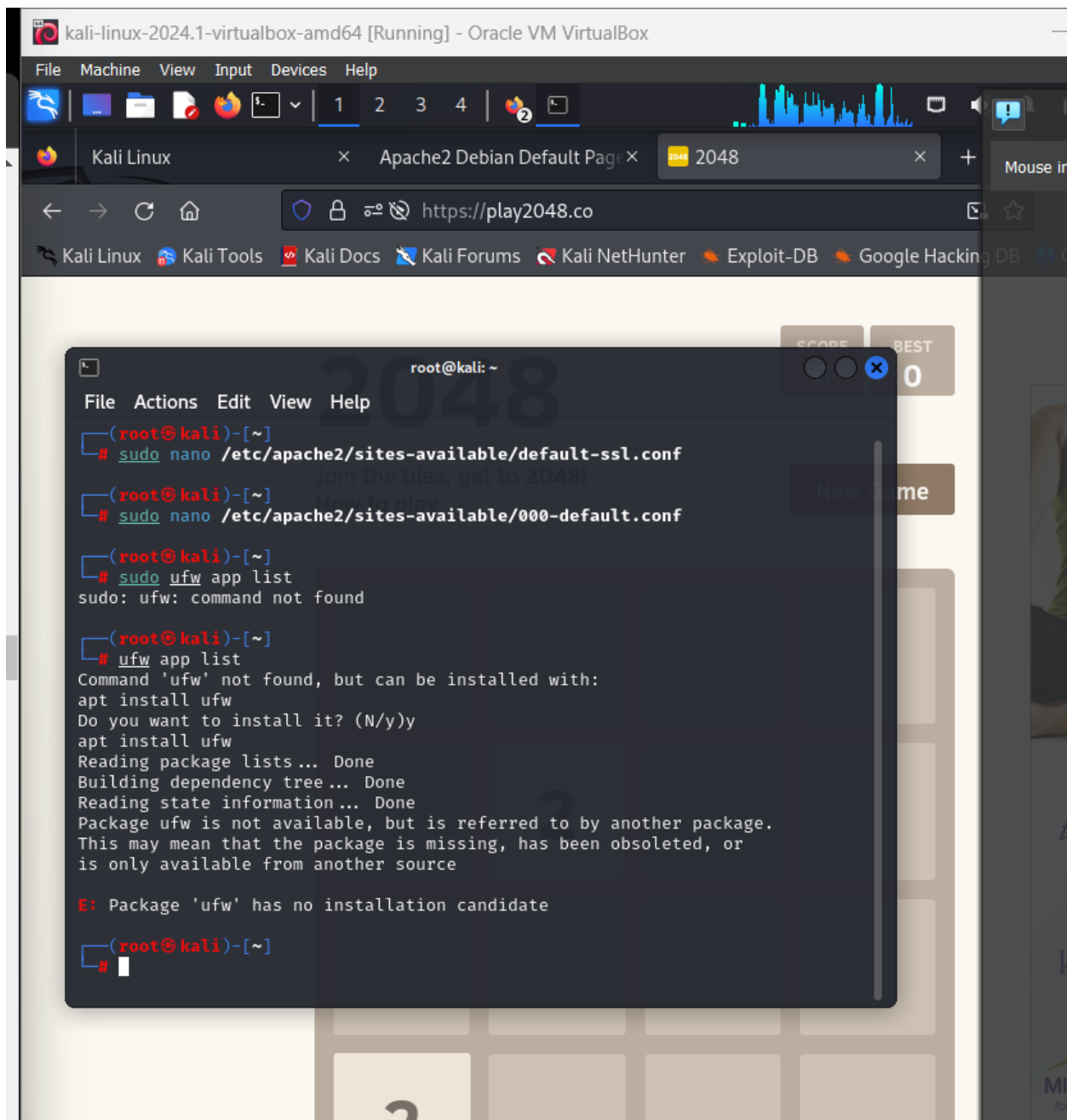
    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined
```

At the bottom of the nano editor, there is a help bar with the following shortcuts:

<code>^G</code> Help	<code>^O</code> Write Out	<code>^W</code> Where Is	<code>^K</code> Cut	<code>^T</code> Execute
<code>^X</code> Exit	<code>^R</code> Read File	<code>^M</code> Replace	<code>^U</code> Paste	<code>^J</code> Justify

The terminal window is overlaid on a background that appears to be a game interface, with a large number '2' visible at the bottom.

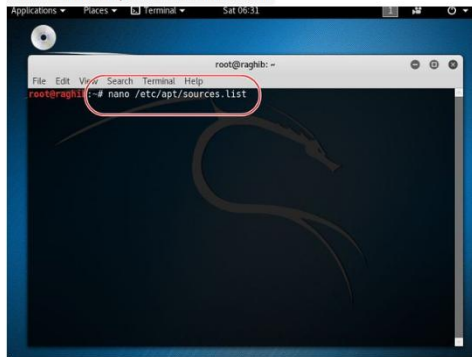
I tried to use the ufw command to configure the firewall but after multiple attempts to edit the linux files and even change to debian and then trying suggested commands for a kali linux OS it led to more errors.



```
root@kali: ~  
File Actions Edit View Help  
(root@kali)-[~]  
# sudo nano /etc/apache2/sites-available/default-ssl.conf  
(root@kali)-[~]  
# sudo nano /etc/apache2/sites-available/000-default.conf  
(root@kali)-[~]  
# sudo ufw app list  
sudo: ufw: command not found  
(root@kali)-[~]  
# ufw app list  
Command 'ufw' not found, but can be installed with:  
apt install ufw  
Do you want to install it? (N/y)y  
apt install ufw  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Package ufw is not available, but is referred to by another package.  
This may mean that the package is missing, has been obsoleted, or  
is only available from another source  
  
E: Package 'ufw' has no installation candidate  
(root@kali)-[~]  
#
```

Open Terminal and type
first you must update repository

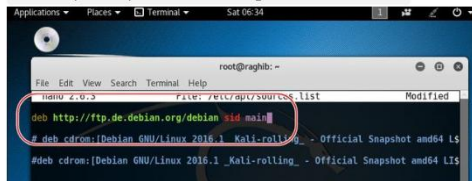
```
#nano /etc/apt/sources.list.txt
```



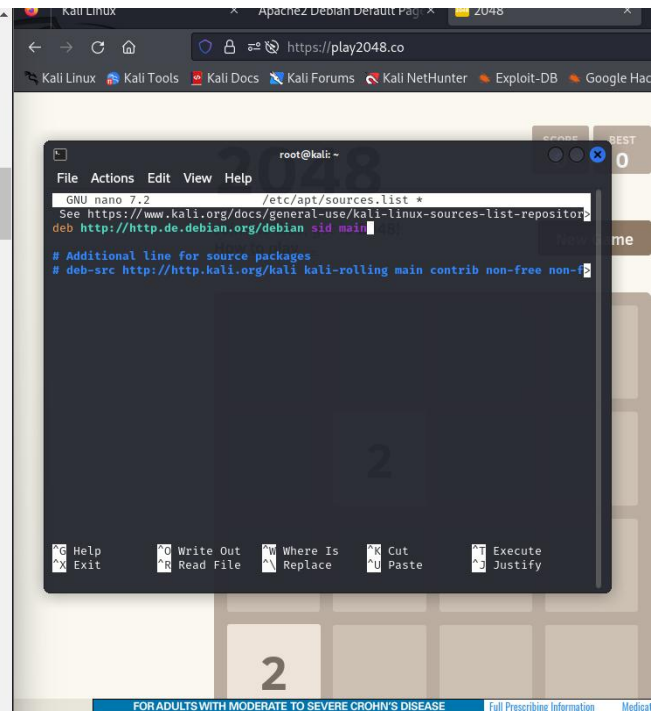
```
root@kali:~# nano /etc/apt/sources.list
```

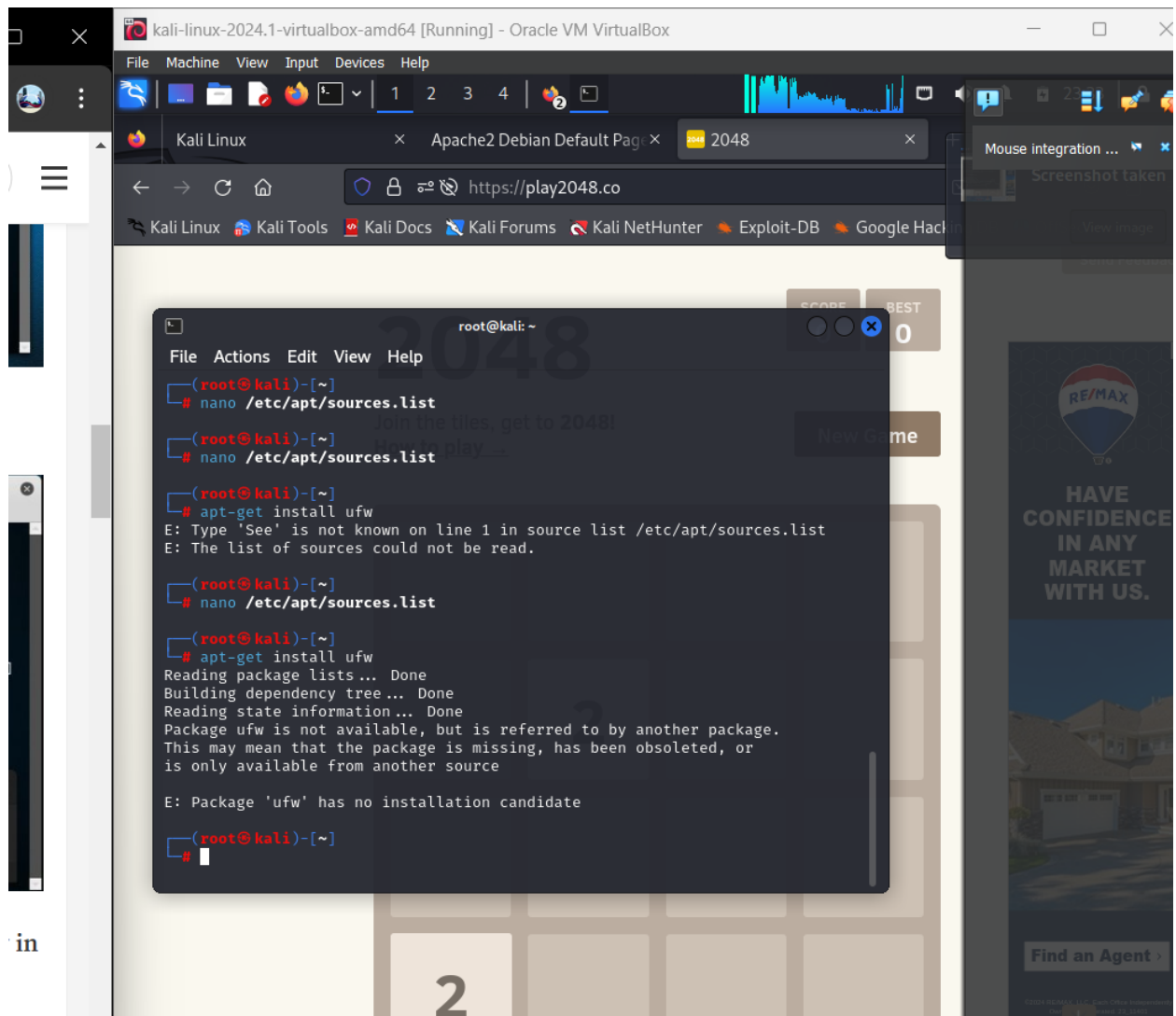
then, write this sourclist.

```
deb http://ftp.de.debian.org/debian sid main
```



```
root@kali:~# nano /etc/apt/sources.list
#deb cdrom:[Debian GNU/Linux 2016.1_Kali-rolling_ - Official Snapshot amd64 L5]
deb http://ftp.de.debian.org/debian sid main
#deb cdrom:[Debian GNU/Linux 2016.1_Kali-rolling_ - Official Snapshot amd64 L15]
```





in